

other hand, Google does include a 'download a pdf' button for all the out of copyright books on its platform.

With publishing partners and university libraries increasingly joining in over the years, the number of books on the Google Books platform has been consistently rising. As of October 2015, there were over 25 million scanned titles in the platform. To put that into context, Google estimates that there are about 130 million distinct titles around the world - and as is evident, they have already supposedly scanned almost 1/5th of them. Receiving its fair share of copyright battles and court cases over the years, Google Books has settled most of them and continues on its efforts to digitise every written word in a book ever.

TensorFlow

TensorFlow is an open-source software library or framework that uses data flow graphs for numerical computation. It can be regarded as a symbolic math library which is utilized to build and train neural networks in order to detect patterns which are in line with how the human brain learns and reasons. Before it went open-source, TensorFlow was developed by the team working on Google Brain while they were researching on deep neural networks.

Ever since TensorFlow became open, there have been quite significant strides in the field as deep learning has helped computers to parse natural language by enabling software to effectively respond to spoken queries. In TensorFlow, a Tensor is a typed multi-dimensional data array on which operations are performed by neural networks.

TPUs

Google I/O hailed the announcement of Google's proprietary Tensor Processing Unit (TPU) which is a custom Application Specific Integrated Circuit (ASIC) designed specially for machine learning. A TPU provides higher IOPS (Input/Output Operations per Second) per watt and is designed for a really high volume of reduced precision computation and they don't have the requisite hardware for texture mapping in comparison to GPUs. TPUs have been developed specifically for TensorFlow.

TPUs have been great at speeding up the process of machine learning which is visible in a recent breakthrough. TPUs were used in the man vs. machine based AlphaGo series (Google DeepMind challenge match) where AI program AlphaGo successfully beat 18-time world champion Lee Sedol